

Francesco Schiavone

Curriculum vitae

Last updated August 1, 2026

Work experience

INFN Research Fellow INFN Sezione di Bari Research topic: <i>Study of cosmic radiation with Fermi-LAT</i>	June 2026 – present Bari, Italy
Peer tutor for the General Physics I and II courses Università degli Studi di Bari “Aldo Moro”	October 2025 – April 2026 Bari, Italy
Peer tutor for the General Physics II course Università degli Studi di Bari “Aldo Moro”	March – December 2024 Bari, Italy

Education

PhD in Astroparticle Physics Università degli Studi di Bari “Aldo Moro” Thesis title: <i>Present and future constraints on axion-like particles from blazars: from current gamma-ray telescopes to the Cherenkov Telescope Array Observatory</i> Supervisors: Prof. Francesco Giordano, Dr. Leonardo Di Venere	February 2023 – May 2026 Bari, Italy
Master’s degree in Theoretical Physics Università degli Studi di Bari “Aldo Moro” Thesis title: <i>Fluxes and signatures of gravitons and axions from the evaporation of primordial black holes</i> Thesis supervisors: Prof. Alessandro Mirizzi, Prof. Daniele Montanino Grade: 110/110 <i>cum laude</i>	September 2018 – April 2021 Bari, Italy
Bachelor’s degree in Physics Università degli Studi di Bari “Aldo Moro” Thesis title: <i>Conservation of parity</i> (original: <i>La conservazione della parità</i>) Thesis supervisor: Prof. Domenico Di Bari Grade: 110/110 <i>cum laude</i>	September 2015 – July 2018 Bari, Italy
High school diploma Liceo Scientifico Statale “Arcangelo Scacchi” Grade: 100/100 <i>cum laude</i>	September 2010 – July 2015 Bari, Italy

Affiliations

MAGIC Collaboration Member	October 2024 – present
COST Action CA21106 Theme: <i>COSMIC WISPerS in the Dark Universe: Theory, astrophysics and experiments</i> Member of Working Groups 3 (WISPs in Astrophysics) and 4 (Direct WISPs searches)	June 2023 – present
Cherenkov Telescope Array Observatory (CTAO) Consortium Member	March 2023 – present

CTAO Large-Sized Telescope (CTAO-LST) Collaboration
Member

March 2023 – present

Istituto Nazionale di Fisica Nucleare (INFN) – Sezione di Bari
Member

March 2023 – present

Activity in international collaborations

LST Cherenkov camera assembly for CTAO South 25 – 28 May 2026
University of Catania (IT)

MAGIC Operator 5 – 29 March 2026 (P287)
Observatorio del Roque de los Muchachos, La Palma (ES)

LST-1 Deputy Shift Leader 15 July – 4 August 2025 (P079)
Observatorio del Roque de los Muchachos, La Palma (ES)

LST-1 Flare Advocate 18 March – 7 April 2025 (P075)
Remote

LST-1 Deputy Flare Advocate 19 November – 11 December 2024 (P071)
Remote

LST-1 Operator 27 February – 19 March 2024 (P062)
Observatorio del Roque de los Muchachos, La Palma (ES)

Lectures and seminars

Seminar for the students of the Multimessenger Astrophysics course 10 November 2025
Università degli Studi di Bari “Aldo Moro”
Title: *Probing dark matter and fundamental physics with Cherenkov telescopes*

Seminar for the students of the High Energy Astrophysics course 26 March 2024
Università degli Studi di Bari “Aldo Moro”
Title: *Basics of optical telescopes*

Invited seminar 12 December 2021
University of Zaragoza (ES)
Title: *Axion-like particles from primordial black holes shining through the Universe*

Virtual Axion Institute 19 July 2021
Online ([YouTube link](#))
Title: *Axion-like particles from PBHs shining through the Universe*

Publications

I am co-author of 25 papers and proceedings, with a h -index of 7 according to [INSPIREHEP](#). The full list of publications is available on [ORCID](#). A selection is given below:

- [1] F. Schiavone, L. Di Venere and F. Giordano. “Projected sensitivity of CTAO to axion-like particles from blazars with a machine learning approach”. *Eur. Phys. J. C* **86**, 639 (2026). doi: [10.1140/epjc/s10052-026-15910-x](#). arXiv: [2512.19259](#) [astro-ph].
- [2] F. Schiavone for the CTAO Consortium. “Dark matter and fundamental physics with the Cherenkov Telescope Array Observatory”. *Il Nuovo Cimento* **49 C**, 7 (2026). doi: [10.1393/ncc/i2026-26007-6](#). arXiv: [2509.24523](#) [astro-ph.HE].

- [3] F. Schiavone, L. Di Venere, F. Giordano for the CTAO Consortium. “A machine learning approach to axion-like particle searches in CTAO observations of blazars”. *Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025)* **501**, 834 (2025). doi: [10.22323/1.501.0834](https://doi.org/10.22323/1.501.0834). arXiv: [2509.24543](https://arxiv.org/abs/2509.24543) [astro-ph].
- [4] F. Schiavone, L. Di Venere, F. Giordano for the CTAO Consortium. “CTAO sensitivity to axion-like particles”. *EPJ Web of Conferences* **319**, 13001 (2025). doi: [10.1051/epjconf/202531913001](https://doi.org/10.1051/epjconf/202531913001).
- [5] F. Schiavone, A. Mirizzi, D. Montanino, F. Capozzi. “Axion-like particles from primordial black holes shining through the Universe”. *JCAP* **08** (2021), p. 063. doi: [10.1088/1475-7516/2021/08/063](https://doi.org/10.1088/1475-7516/2021/08/063). arXiv: [2107.03420](https://arxiv.org/abs/2107.03420) [hep-ph].

Contributions to conferences and workshops

Selected list of recent contributions:

- TeVPA 2025** 6 November 2025
Fundación ADEIT, Valencia (ES)
Talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*
- 20th Patras Workshop on Axions, WIMPs and WISPs** 22–26 September 2025
IACTEC, La Laguna, Tenerife (ES)
Talk/poster: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*
- 3rd General Meeting of COST Action COSMIC WISPerS (CA21106)** 9 – 12 September 2025
University of Sofia (BG)
Invited talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*
- Extragalactic jets at all scales: a Cretan view** 25 – 29 August 2025
Aquila Atlantis Hotel, Heraklion (GR)
Poster/talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*
- 39th International Cosmic Ray Conference (ICRC 2025)** 14 – 24 July 2025
International Conference Centre (CICG), Geneva (CH)
Poster: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach* (first author, presented by L. Di Venere)
- Incontri di Fisica delle Alte Energie (IFAE) 2025** 10 April 2025
Manifattura Tabacchi, Cagliari (IT)
Talk: *Materia oscura e fisica fondamentale con il Cherenkov Telescope Array Observatory* (in Italian)

Organization of conferences and workshops

As part of Local Organizing Committee:

- 2nd VHEgam Meeting** 26 May 2025
Centro Polifunzionale Studenti, Bari (IT)